

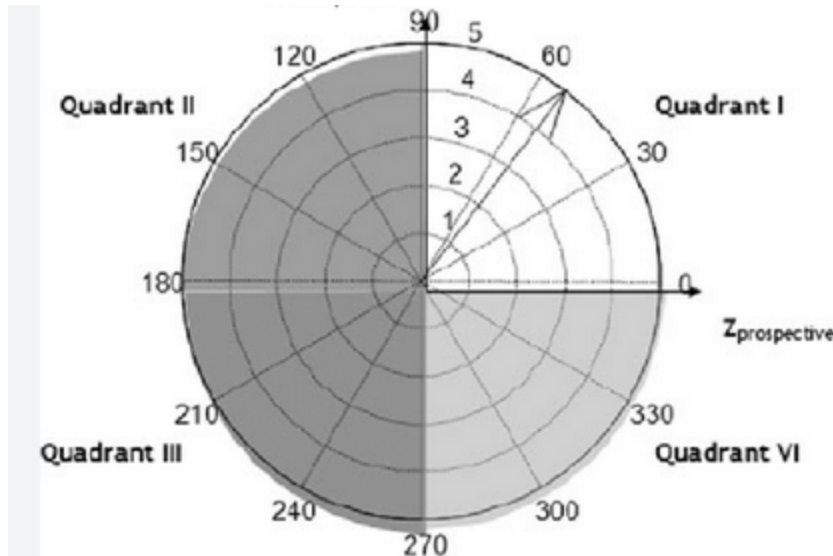
popquiz circular motion, adding vectors

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:56pm

Quiz Instructions

Know your quadrant and polar graph



Question 1 10 pts

 In the third quadrant, what is the sign of both the x-component and y-component of a vector?

- A) Both negative
- B) Both positive
- C) X-component negative, y-component positive
- D) X-component positive, y-component negative



x-component is positive, y-component negative



Both positive



x-component is negative, y-component positive



Both negative



Question 2 10 pts



An old 33 rpm vinyl disk has a radius of 15cm (0.15m). rpm means revolutions per minute.

what is the speed in m/s of a point located at the out edge of the disk?



31m/s



0.015m/s



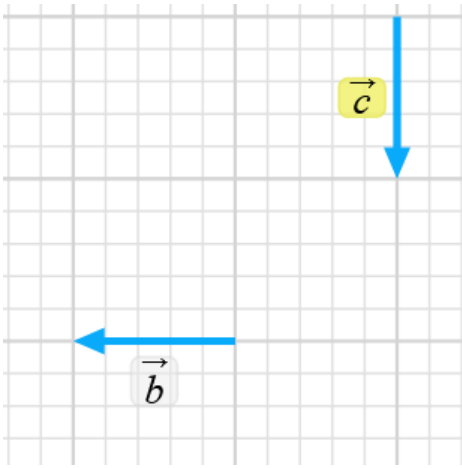
0.52m/s



0.8m/s



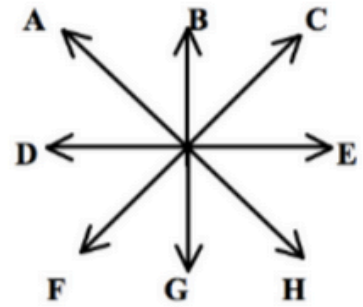
Question 3 10 pts



Here are 2 vectors $\vec{b}$ and $\vec{c}$. Which of the following vectors shows the sum of the 2 vectors.

hint: Use the tail to head method as discussed

- vector A
- vector B
- vector C
- vector D
- vector E
- vector F
- vector G
- vector H


☐

D

☐

E

☐

C

☐

H

☐

G

☐

F

☐

B

☐

A

☐

Question 4 10 pts

Consider 2 vectors. $A(-2,3)$ and $B(5,-4)$.

Use algebra to find the components of the vector sum C

☐

$C(3,-1)$

☐

$C(3,1)$

☐

$C(-7,7)$

☐

$C(7,7)$

☐

Question 5 10 pts

Trace the vector components in a X,Y system. Trace the vector C . Its component is the head. The vector C is

☐

in the 4th quadrant

☐

in the third quadrant.

☐

in the second quadrant

☐

in the first quadrant

☐

Question 6 10 pts

The angle of the vector with $+x$ axis is :

(CW is negative and CCW is positive)

☐

- 20 degrees

☐

110 degrees

☐

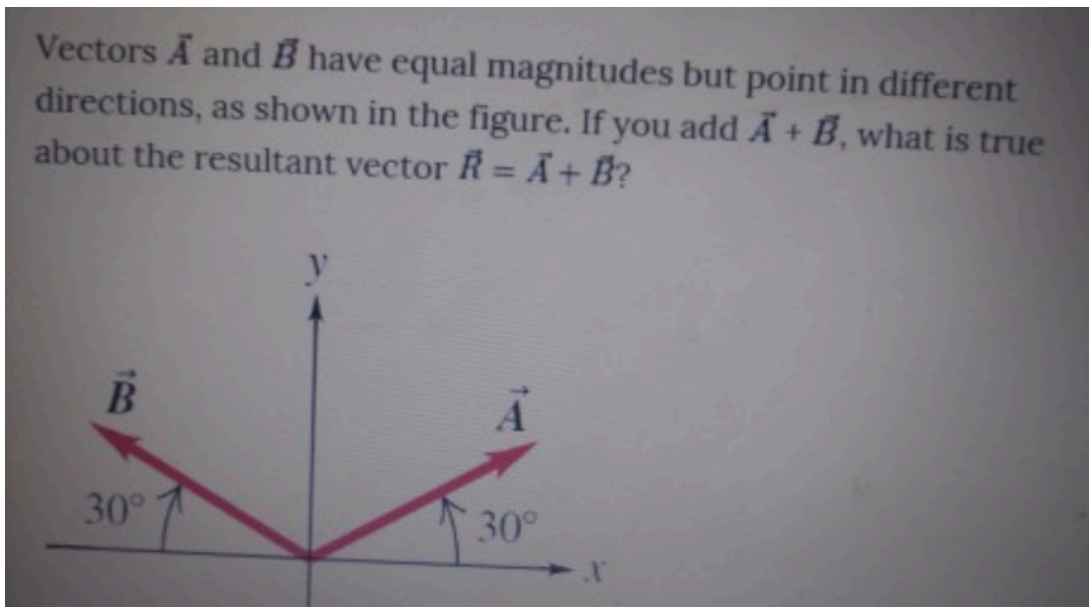
-80 degrees

☐

200 degrees

☐

Question 7 10 pts


☐

The resultant is along +yaxis and has a magnitude larger than the magnitude of A or B

☐

The resultant is along the x-axis and is equal to the magnitude of A or B

☐

The resultant is along +yaxis and has a magnitude equal to the magnitude of A or B

☐

The resultant is along the negative y-axis and has a magnitude larger than the magnitude of A or B

⋮

Question 8 10 pts



Which of the following best represent vector sum $\vec{B} + \vec{A}$?

a)



b)



c)



d)


☐

A

☐

B

☐

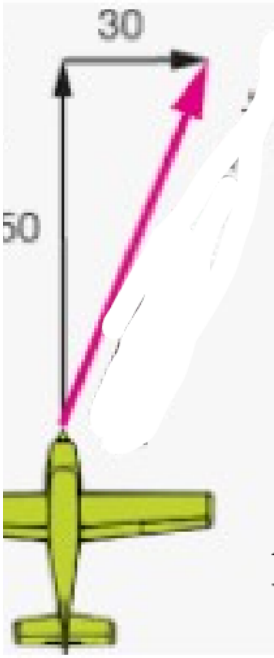
C

☐

D

⋮

Question 9 10 pts



This is not a picture on scale. The airplane is going at 50km/s @ North relative to wind. And the wind is going at 30km/s @ East relative to ground. The velocity of the plane relative of the ground is in pink. you need to solve for this velocity relative to ground. What is the velocity? angle is from East. CCW.

East direction is like the positive x-axis. you want to use the polar graph. Round the angle

- ☐ 58km/s @ 30 above +x-axis (east)
- ☐ 58km/s @ 120 above +x-axis (east)
- ☐ 38km/s @ 60 above +x-axis (east)
- ☐ 58km/s @ 60 above +x-axis (east)



Question 10 10 pts

A man walks 15 m to the east then, at 25 m to the west. At the end of his walk relative to his starting point, he is

- a) 40 m to the east c) 25 m to the North
- b) 40 m to the west d) 10 m to the west

- ☐ D
- ☐ B



A



C

Not saved

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